

PARTICIPANT WORKBOOK

The EU Climate Machine

Exercises, worksheets, the formula sheet and an annotated source list — yours to keep and keep using.

Name: _____ Session date: _____

Edition: August 2026



How to use this workbook

Each module has three parts here: a one-paragraph recap of the mechanism, a worksheet you fill in during the exercise, and space for your own notes. The rules that make the day work: write your **prediction before you run any model** (the learning lives in the gap between the two), and treat every model as what it is — a stylised teaching instrument that reproduces the logic and order of magnitude of published analyses, not a forecast and not a citable source. The formula sheet and source list at the back are the exception: those are built for reuse.

The day at a glance

Time	Session
09:00	Welcome and framing — the July 2026 package
09:30	Module 1 · The carbon pricing endgame
11:15	Module 2 · Why European electricity costs what it costs
13:30	Module 3 · Electrification and the 2040 framework
15:15	Module 4 · Stress tests — politics and nature
16:15	Synthesis — your watchlist

Module 1 · The carbon pricing endgame

The mechanism in one paragraph. The ETS is a cumulative budget: a shrinking cap plus banking means the allowance price is set by intertemporal arbitrage — it rises at roughly the discount rate while the bank is positive, and the whole path is pinned by the one starting price that makes cumulative emissions exhaust the cumulative cap plus the bank. Abatement gets more expensive as the easy tonnes run out (a convex MACC saturating at ≈81% of baseline), so a price-only system spikes as the cap approaches zero. Permanent removals act as a price-responsive backstop — but only up to the volume policy admits.

Exercise 1 · Price the July 2026 package

Step 1 — map the real package onto the model's dials:

Package element	Which dial(s)?	Direction of price effect?
LRF slowed to 3.7% (2031–35), 1.7% (2036–40)		
250 Mt permanent removals, on top of the cap (Art. 9c)		
MSR intake halved 24% → 12%		
≈260 Mt international credits from 2036 (2033 review)		

Step 2 — predict before you run (base case: €195 in 2030, €336 in 2040):

Your predicted 2030 price	Your predicted 2040 price
Prediction	
Model output	
Why the difference?	

Step 4 — your team's one sentence (who should be nervous, and why):

Model reference card

Parameter	Default	Parameter	Default
Cap 2025 / net-zero year	1,340 Mt / 2040	Banked headroom	1,500 Mt
Hotelling rate r	5.6%/yr	Abatement ceiling $\phi_{\max}$	0.81
CDR floor cost	€70/t	CDR admitted (max)	90 Mt/yr from 2031
Integrity haircut	10%	Capacity ceiling	165 Mt/yr by 2050

Module 2 · Why European electricity costs what it costs

The mechanism in one paragraph. Day-ahead markets are uniform-price auctions: every plant bids $\approx$ its short-run marginal cost, bids stack cheapest-first, and the last plant needed sets the price paid to all. The average cost of the fleet is irrelevant; only the marginal machine matters — and in Europe that machine is a gas plant in over half of all hours. Feed the same auction US or Chinese inputs and it produces US or Chinese prices: the gap is an input gap, not a mechanism gap.

Exercise 2 · Build the merit order

Parameter card — fuel €/MWh thermal: gas 38 · hard coal 13 · lignite 5. CO_2 : €70/t. Emission factors ($\text{tCO}_2/\text{MWhth}$): gas 0.202 · coal 0.34 · lignite 0.40. Formula: $\text{MC} = \text{fuel}/\eta + (\text{EF}/\eta) \cdot \text{CO}_2 + \text{O\&M}$.

Plant (capacity)	η	O&M;	Your MC computation	MC €/MWh
Nuclear (78 GW)	—	$\approx$ €12 all-in		
Hard coal (38 GW)	0.42	€3		
Lignite (28 GW)	0.40	€12		
CCGT gas (145 GW)	0.56	€2		
OCGT peaker (45 GW)	0.38	€3		

Now stack cheapest-first (wind & solar bid €0: 105 GW running) and clear the market at **350 GW** demand. Marginal plant: _____ Clearing price: _____

Re-run	Marginal plant	Clearing price
August 2022: gas €235/MWhth, nuclear derated		
US inputs: gas $\approx$ €11, no carbon price		

Your 30-second answer: “why does a 70%-renewable grid still price at gas?”

Module 3 · Electrification and the 2040 framework

The mechanism in one paragraph. Final demand in buildings heat, road transport and industry heat splits into adoption tranches ordered by barrier-adjusted switching cost; a carbon price P electrifies every tranche cheaper than P . Price alone reaches the 46%-by-2040 target at $\approx \text{€}166/\text{t}$ — $3.7\times$ the $\text{€}45/\text{t}$ at which ETS2 releases extra allowances, so that route is institutionally foreclosed. With the demand-side package (price-ratio reform, building and vehicle standards, CCfDs) the same target clears at $\approx \text{€}55/\text{t}$. The $\text{€}111/\text{t}$ gap is the shadow value of demand-side policy. Separately: because electric devices need $\sim 3\times$ less input energy, every switch shrinks the final-energy denominator — the 46% metric lags real device adoption badly.

Exercise 3 · Pick your mix on the iso-target frontier

Precomputed from the identity (COP 3.0, boiler η 0.9, ICE÷EV 3.3, heat 2,500 TWh, cars 1,700 TWh, total 8,100 TWh). Today: heat pumps $\approx 13\%$, EV $\approx 4\%$ of car-km.

If heat pumps reach...	EV share needed (no other help)	EV share needed (25% of other fossil switched)
50%	>100% — unreachable	$\approx 79\%$
60%	$\approx 101\%$ — unreachable	$\approx 63\%$
70%	$\approx 85\%$	$\approx 48\%$
75% (Liebreich's point)	$\approx 78\%$	$\approx 40\%$
80%	$\approx 70\%$	$\approx 32\%$
90%	$\approx 54\%$	$\approx 16\%$

Your 2040 mix	Heat-pump share	EV share	Other sectors switched	Clears the frontier?
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Team consensus

Which single policy from the package does your mix lean on hardest?

The fleet-life check: what would 2027 sales have to look like for your mix?

Module 4 · Stress tests — politics and nature

The two identities. Politics: under a binding banked cap, cumulative emissions $\approx$ cumulative cap – final bank — so demands that move the cap or add fungible supply cost tonnes directly, while free-allocation demands mostly redistribute rent inside the cap. Nature: the 2040 target is net, so every tonne the land sink fails to absorb converts one-for-one into deeper cuts elsewhere; the wildfire cohort model turns burnt hectares into that shortfall.

Debate prompt (if run)

Your demand	Cost/gain in the central reading	Your position (attack / defend)
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Your argument — tonnes only, no adjectives:

Synthesis · Your watchlist

Pick three of the day's five numbers plus two indicators specific to your organisation. Name the source you will actually check, and the threshold that would make you act.

Indicator	Source you will check	Check every	Act if...
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1.

2.

3.

4. (yours)

5. (yours)

The five candidates: ① EUA price vs the $\approx$ €195-by-2030 anchor · ② your member state's electricity-to-gas retail price ratio vs the $\leq 2.5 / \leq 2$ KPI · ③ FID-backed CDR pipeline vs the ~ 48 Mt/yr 2040 need · ④ LULUCF sink outturn vs 310 Mt by 2030 · ⑤ the 2033 international-credit review (fail $\Rightarrow$ LRF reverts to 2.7%).

The formula sheet

Model	The formula(s)	Reading
ETS price path	$\text{price}(t) = P_0 \cdot (1+r)^{(t-2025)}$; P_0 solves $\Sigma \text{net} = \Sigma \text{cap} + \text{bank}$	One number pins the path; tighten anything cumulative and today's price moves.
Abatement	$a\text{Conv}(P) = \text{BAU} \cdot \phi_{\text{max}} \cdot P/(P + 235)$	Convex MACC saturating at 81% — the endgame spike is the last 19%.
CDR valve	$\text{removals} = \min(\text{supply}(P), \text{capacity}, \text{admitted}) \cdot (1 - \text{haircut})$	Three constraints; the admitted volume is the policy dial.
Merit order	$\text{MC} = \text{fuel}/\eta + (\text{EF}/\eta) \cdot \text{CO}_2 + \text{O\&M}$; price = MC of last plant needed	Average fleet cost is irrelevant; only the marginal machine matters.
Electrification LCM	$\text{switch_cost}(f) = \text{base} + \text{spread} \cdot f^\gamma$; electrify every tranche $< P$	Measures shrink the spread and base — that is the €111/t.
Final-energy identity	$\text{rate} \approx (1/\text{COP}) \div (1/\text{COP} + 1/\eta)$ in the two-device toy	The denominator shrinks as you succeed; shares lag devices.
Wishlist accounting	$\Sigma \text{emissions} \approx \Sigma \text{cap} - \text{final bank}$	Cap/supply demands cost tonnes; free-allocation demands move rent.
Wildfire cohorts	$\text{loss} = \text{combustion} + \text{decomposition} + \text{foregone regrowth} - \text{embedded baseline}$	Only the EXCESS over what projections assume counts against the target.

The numbers worth memorising

≈€73/MWh	EU–US wholesale gap at the marginal generator: ≈€48 fuel + €25 carbon
€166 / €55 / €111	Price-only vs with-package carbon price for 46% electrification; the gap = shadow value of demand-side policy
€45/t	The ETS2 trigger that forecloses the price-only route
≈54×	Scale-up from today's FID-backed CDR pipeline to the 2050 safety-valve need
+1,294 Mt	Central cost of the ten softening demands to 2040 (≈11.8 years of EU progress)
47 Mt = 44%	Default 2040 wildfire sink loss = share of the planned sink improvement it takes back

Annotated source list

COM(2026) 616 + SWD(2026) 616 (impact assessment, 5 parts, 958 pp).

The ETS review package and its evidence base — LRF, Removals Authority, credits, MSR, CBAM timing. The IA's own uncertainty admissions (e.g. costs and benefits of the preferred option could not be monetised) are as instructive as its numbers.

COM(2026) 595 — Electrification Action Plan.

The 46% indicative target, the price-ratio KPI ($\leq 2.5 / \leq 2$), and the four pillars; to be enshrined via the post-2030 Energy Union package.

Sultani et al., “Sequencing Carbon Dioxide Removal into the EU ETS”, Joule (2026); Ariadne safety-valve dossier (PIK).

The published basis for Module 1: price paths ($\approx \text{€}203/\text{€}353$), the 68–86 Mt/yr valve, and the three-stage sequencing proposal.

JRC134300 (2023); ACER market monitoring; Draghi report (2024).

Gas price-setting shares, the 2021–22 crisis anatomy, and the converging prescription: less gas at the margin, keep marginal pricing.

Eurostat nrg_pc_204/205; SMARD/EPEX; EIA.

Retail price stacks and the wholesale history behind Module 2's charts.

Günther, Pahle et al., Climate Policy (2025); Ariadne ETS2 analyses.

The published price envelope (71–261 €/t for 2030) that brackets the workshop's least-cost model.

Liebreich via Euractiv (July 2026).

The final-energy fallacy argument and the $\frac{3}{4} \cdot \frac{4}{5}$ arithmetic Module 3 reproduces.

Copernicus EFFIS annual reports (JRC); Reg. (EU) 2023/839 (LULUCF).

Burnt-area record and the 310 Mt sink target behind Module 4b.

Cames et al. (2016), “How additional is the CDM?”

The ~85% non-additionality finding behind the international-credit default in the wishlist model.

EHPA market reports; JRC144191.

Heat-pump sales collapse of 2024 and the price-ratio evidence behind the cheapest lever.

A closing note on epistemics

Everything you ran today was a stylised teaching instrument: built to reproduce the logic and order of magnitude of the analyses above, and honest about what it leaves out. Cite the sources, not the toys — and when someone shows you a single confident number where you have learned to expect a range, ask to see their sliders.

Notes
